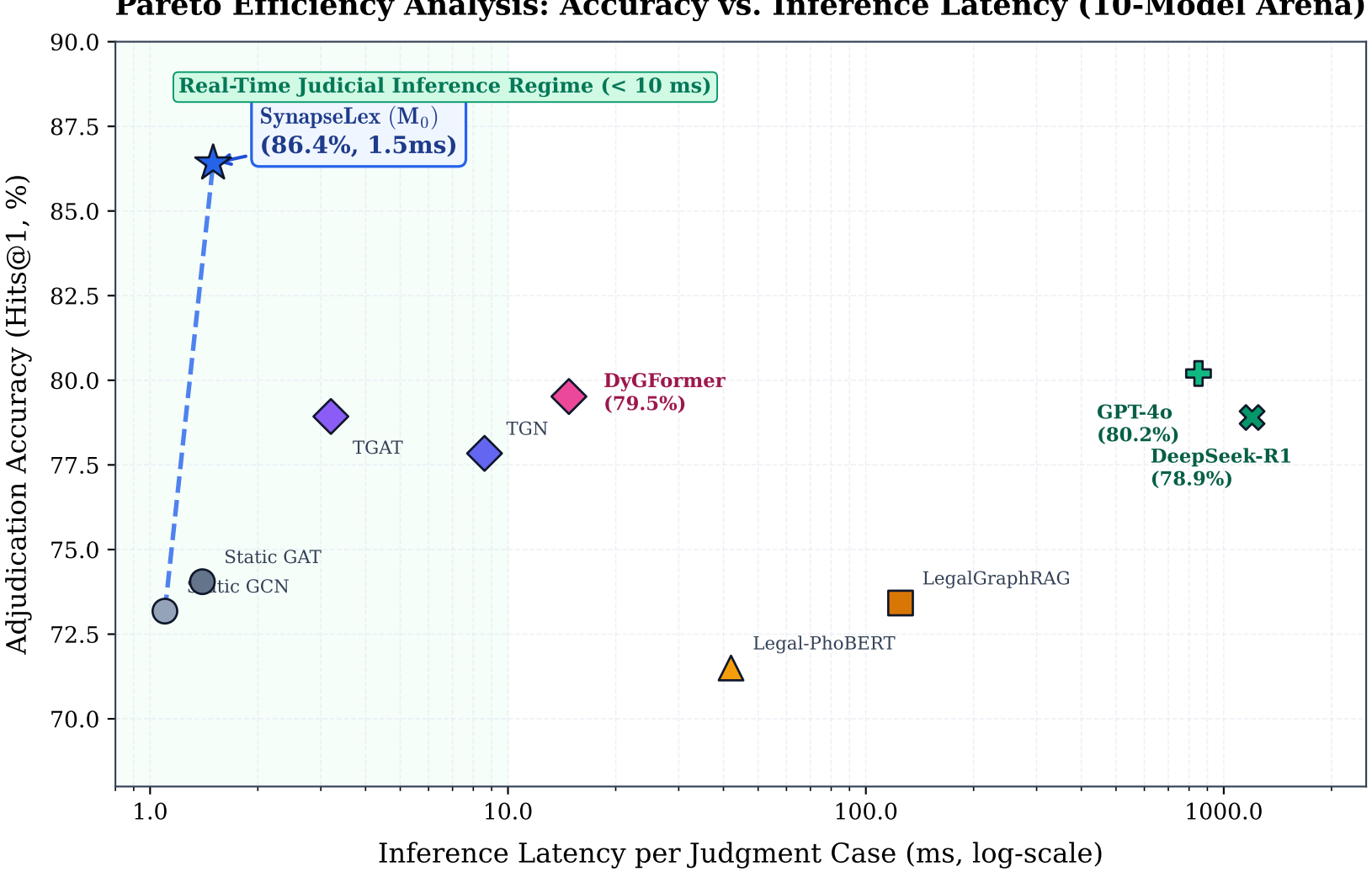
Pareto Efficiency Analysis: Accuracy vs. Inference Latency (10-Model Arena)
Real-Time Judicial Inference Regime (< 10 ms)
SynapseLex ($M_0$)
(86.4%, 1.5ms)
DyGFormer
(79.5%)
TGAT
TGN
GPT-4o
(80.2%)
DeepSeek-R1
(78.9%)
Static GAT
Static GCN
LegalGraphRAG
Legal-PhoBERT
Adjudication Accuracy (Hits@1, %)
Inference Latency per Judgment Case (ms, log-scale)
90.0
87.5
85.0
82.5
80.0
77.5
75.0
72.5
70.0
1.0
10.0
100.0
1000.0